Residue-pair PMFs (6×6)

each mini panel: x 0.3–1.8 nm, y −1.75…2.5 kcal/mol

Three-residue window PMFs

same axes; rows/cols = window on pep 1 / pep 2

LIVA–LIVA fragment PMF

mean of halves ± |PMF₁−PMF₂|/2

H-bond occupancy (LIVA), by direction

rows = donor, cols = acceptor; per contact frame, mean ± |h₁−h₂|/2

H-bond network

filled = pep 1, open = pep 2; arrow = donor N–H → acceptor O

pep 1 donates (system colour) pep 2 donates (gray)

pep1–pep2 angle

mean of half densities ± |h₁−h₂|/2

End-to-end distance (pep 1)

mean of half densities ± |h₁−h₂|/2

2D PMF: angle × end-to-end, pep A

pep A = pep 1, :2@CA–:7@CA; contact frames, halves pooled

2D PMF: angle × end-to-end, pep B

pep B = pep 2, :10@CA–:15@CA; same frames and bins

2D PMF: average of A and B

(F_A + F_B)/2 per bin; each panel 0 at its lowest bin

LIVAGE (E⁻/E⁻)

GLU/GLU

−1.31 ± 0.10 at 0.42 nm

mean 106.6

mean 13.0

LIVAGE (E⁻/E⁰)

GLU/GLH

−1.08 ± 0.04 at 0.42 nm

mean 98.5

mean 12.9

LIVAGE (E⁰/E⁰)

GLH/GLH

−0.96 ± 0.15 at 0.43 nm

mean 99.1

mean 12.2

LIVAGG (G⁰/G⁰)

GLY/GLY

−1.13 ± 0.13 at 0.42 nm

mean 105.1

mean 12.7

LIVAGH (H⁺/H⁺)

HIP/HIP

−0.85 ± 0.09 at 0.42 nm

mean 105.0

mean 12.5

LIVAGK (K⁺/K⁺)

LYS/LYS

−1.13 ± 0.03 at 0.52 nm

mean 105.9

mean 12.8

LIVAGQ (Q⁰/Q⁰)

GLN/GLN

−1.04 ± 0.02 at 0.52 nm

mean 100.7

mean 12.3

LIVAGR (R⁺/R⁺)

ARG/ARG

−0.93 ± 0.05 at 0.42 nm

mean 105.2

mean 12.4

PMF (kcal/mol)

COM distance (nm)

density

angle (deg)

end-to-end distance (Å)

end-to-end Cα-Cα (Å)

pep1–pep2 angle (deg)

pep 1 → pep 2 occupancy (each row in its system colour)

pep 2 → pep 1 occupancy

2D PMF (kcal/mol), 0 = lowest bin; lines = contours 1, 2 (each row in its system colour)